

# Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.30)

## Metric

$$ds^2 = -2 \left( A - \frac{b_x^2 + b_y^2}{P} \right) du^2 - 2 du dv + \frac{4b_x}{P} du dx + \frac{4b_y}{P} du dy + \frac{2}{P} (dx^2 + dy^2).$$

## Coordinates

$$(x^\mu) = (u, v, x, y)$$

## Metric Functions

$$K = K(u),$$

$$H = H(u, x, y),$$

$$Q = 1 + \epsilon(x^2 + y^2),$$

$$P = k^2 Q^2,$$

$$A = \epsilon v^2 k^2 + lv + H,$$

$$b_x = b_x(u, x, y),$$

$$b_y = b_y(u, x, y).$$

## Null Tetrad

*Covariant components*

$$l = (1) du$$

$$n = \left( \epsilon v^2 k^2 + lv - \frac{b_x(u, x, y)^2}{(\epsilon^2 x^4 + 2\epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2\epsilon x^2 + 2\epsilon y^2 + 1)k^2} - \frac{b_y(u, x, y)^2}{(\epsilon^2 x^4 + 2\epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2\epsilon x^2 + 2\epsilon y^2 + 1)k^2} + H \right) du + (1) dv \\ + \left( -\frac{2b_x(u, x, y)}{(\epsilon^2 x^4 + 2\epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2\epsilon x^2 + 2\epsilon y^2 + 1)k^2} \right) dx + \left( -\frac{2b_y(u, x, y)}{(\epsilon^2 x^4 + 2\epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2\epsilon x^2 + 2\epsilon y^2 + 1)k^2} \right) dy$$

$$m = \left( \sqrt{\frac{1}{(\epsilon^2 x^4 + 2 \epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2 \epsilon x^2 + 2 \epsilon y^2 + 1)k^2}} \right) dx + \left( i \sqrt{\frac{1}{(\epsilon^2 x^4 + 2 \epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2 \epsilon x^2 + 2 \epsilon y^2 + 1)k^2}} \right) dy$$

$$\bar{m} = \left( \sqrt{\frac{1}{(\epsilon^2 x^4 + 2 \epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2 \epsilon x^2 + 2 \epsilon y^2 + 1)k^2}} \right) dx + \left( -i \sqrt{\frac{1}{(\epsilon^2 x^4 + 2 \epsilon^2 x^2 y^2 + \epsilon^2 y^4 + 2 \epsilon x^2 + 2 \epsilon y^2 + 1)k^2}} \right) dy$$

*Contravariant components*

$$l = (-1) dv$$

$$n = (-1) du + \left( \frac{\epsilon^3 v^2 x^4 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^3 v^2 x^2 y^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \right. \\ + \frac{\epsilon^3 v^2 y^4 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon^2 l v x^4 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{2 \epsilon^2 l v x^2 y^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon^2 l v y^4 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{2 \epsilon^2 v^2 x^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^2 v^2 y^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{\epsilon^2 x^4 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^2 x^2 y^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{\epsilon^2 y^4 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon l v x^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{2 \epsilon l v y^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon v^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{2 \epsilon x^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon y^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ + \frac{l v k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ \left. - \frac{b_x(u, x, y)^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} - \frac{b_y(u, x, y)^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \right) dv$$

$$m = \left( \frac{b_x(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i b_y(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} \right) dv + \left( \frac{1}{2} |\epsilon x^2 k + \epsilon y^2 k + k| \right) dx \\ + \left( \frac{i \epsilon^2 x^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon^2 x^2 y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon^2 y^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon x^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \right) dy$$

$$\begin{aligned}\bar{m} = & \left( \frac{b_x(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i b_y(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} \right) dv + \left( \frac{1}{2} |\epsilon x^2 k + \epsilon y^2 k + k| \right) dx \\ & + \left( -\frac{i \epsilon^2 x^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 x^2 y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon x^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \right) dy\end{aligned}$$

## Directional Derivatives

$$D = -\partial_v$$

$$\begin{aligned}\Delta = & -\partial_u + \frac{\epsilon^3 v^2 x^4 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^3 v^2 x^2 y^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{\epsilon^3 v^2 y^4 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon^2 l v x^4 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{2 \epsilon^2 l v x^2 y^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon^2 l v y^4 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{2 \epsilon^2 v^2 x^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^2 v^2 y^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{\epsilon^2 x^4 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon^2 x^2 y^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{\epsilon^2 y^4 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon l v x^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{2 \epsilon l v y^2 k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{\epsilon v^2 k^4}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{2 \epsilon x^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{2 \epsilon y^2 H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & + \frac{l v k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} + \frac{H k^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \\ & - \frac{b_x(u, x, y)^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} - \frac{b_y(u, x, y)^2}{\epsilon^2 x^4 k^2 + 2 \epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2 \epsilon x^2 k^2 + 2 \epsilon y^2 k^2 + k^2} \partial_v\end{aligned}$$

$$\begin{aligned}\delta = & \frac{b_x(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i b_y(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} \partial_v + \frac{1}{2} |\epsilon x^2 k + \epsilon y^2 k + k| \partial_x + \frac{i \epsilon^2 x^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \\ & + \frac{i \epsilon^2 x^2 y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon^2 y^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon x^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \partial_y\end{aligned}$$

$$\bar{\delta} = \frac{b_x(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i b_y(u, x, y)}{|\epsilon x^2 k + \epsilon y^2 k + k|} \partial_v + \frac{1}{2} |\epsilon x^2 k + \epsilon y^2 k + k| \partial_x - \frac{i \epsilon^2 x^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{i \epsilon^2 x^2 y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^4 k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon x^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y^2 k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i k^2}{2 |\epsilon x^2 k + \epsilon y^2 k + k|} \partial_y$$

## Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i \epsilon x^2 k \frac{\partial}{\partial y} b_x(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} - \frac{i \epsilon y^2 k \frac{\partial}{\partial y} b_x(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} + \frac{i \epsilon x^2 k \frac{\partial}{\partial x} b_y(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} + \frac{i \epsilon y^2 k \frac{\partial}{\partial x} b_y(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} + \frac{2i \epsilon y b_x(u, x, y) k}{\epsilon x^2 k + \epsilon y^2 k + k} \\ - \frac{2i \epsilon x b_y(u, x, y) k}{\epsilon x^2 k + \epsilon y^2 k + k} + \frac{\epsilon x^2 \frac{\partial}{\partial u} k}{\epsilon x^2 k + \epsilon y^2 k + k} + \frac{\epsilon y^2 \frac{\partial}{\partial u} k}{\epsilon x^2 k + \epsilon y^2 k + k} - \frac{i k \frac{\partial}{\partial y} b_x(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} + \frac{i k \frac{\partial}{\partial x} b_y(u, x, y)}{2 (\epsilon x^2 k + \epsilon y^2 k + k)} + \frac{\frac{\partial}{\partial u} k}{\epsilon x^2 k + \epsilon y^2 k + k}$$

$$\lambda = 0$$

$$\nu = \frac{\epsilon^3 x^6 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon^3 x^4 y^2 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon^3 x^2 y^4 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^3 y^6 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{i \epsilon^3 x^6 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{3i \epsilon^3 x^4 y^2 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{3i \epsilon^3 x^2 y^4 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^3 y^6 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ + \frac{3 \epsilon^2 x^4 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon^2 x^2 y^2 k^3 \frac{\partial}{\partial x} H}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon^2 y^4 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{3i \epsilon^2 x^4 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{3i \epsilon^2 x^2 y^2 k^3 \frac{\partial}{\partial y} H}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{3i \epsilon^2 y^4 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{2 \epsilon^2 v x^2 b_x(u, x, y) k^3}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{2 \epsilon^2 v y^2 b_x(u, x, y) k^3}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{2i \epsilon^2 v x^2 b_y(u, x, y) k^3}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{2i \epsilon^2 v y^2 b_y(u, x, y) k^3}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon x^2 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{3 \epsilon y^2 k^3 \frac{\partial}{\partial x} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{3i \epsilon x^2 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{3i \epsilon y^2 k^3 \frac{\partial}{\partial y} H}{2 (\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon l x^2 b_x(u, x, y) k}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon l y^2 b_x(u, x, y) k}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{i \epsilon l x^2 b_y(u, x, y) k}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon l y^2 b_y(u, x, y) k}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{2 \epsilon v b_x(u, x, y) k^3}{(\epsilon x^2 k + \epsilon y^2 k + k) |\epsilon x^2 k + \epsilon y^2 k + k|} - \dots$$

$$\alpha = \frac{\epsilon^2 x^3 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 x^2 y k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x y^2 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^3 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon x k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|}$$

$$\beta = -\frac{\epsilon^2 x^3 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 x^2 y k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x y^2 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^3 k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon x k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\begin{aligned} \gamma = & -\frac{\epsilon^2 v x^2 k^2}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{\epsilon^2 v y^2 k^2}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{\epsilon l x^2}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon l y^2}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon v k^2}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{i \epsilon x^2 \frac{\partial}{\partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{i \epsilon y^2 \frac{\partial}{\partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} \\ & + \frac{i \epsilon x^2 \frac{\partial}{\partial x} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{i \epsilon y^2 \frac{\partial}{\partial x} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{i \epsilon y b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{i \epsilon x b_y(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{l}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{i \frac{\partial}{\partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{i \frac{\partial}{\partial x} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} \end{aligned}$$

## Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \epsilon k^2$$



$$\begin{aligned}
\Phi_{22} = & \frac{\epsilon^4 x^8 k^4 \frac{\partial^2}{(\partial x)^2} H}{4(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} + \frac{\epsilon^4 x^6 y^2 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{3\epsilon^4 x^4 y^4 k^4 \frac{\partial^2}{(\partial x)^2} H}{2(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} + \frac{\epsilon^4 x^2 y^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{\epsilon^4 y^8 k^4 \frac{\partial^2}{(\partial x)^2} H}{4(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} + \frac{\epsilon^4 x^8 k^4 \frac{\partial^2}{(\partial y)^2} H}{4(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} \\
& + \frac{\epsilon^4 x^6 y^2 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{3\epsilon^4 x^4 y^4 k^4 \frac{\partial^2}{(\partial y)^2} H}{2(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} \\
& + \frac{\epsilon^4 x^2 y^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{\epsilon^4 y^8 k^4 \frac{\partial^2}{(\partial y)^2} H}{4(\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2)} \\
& + \frac{\epsilon^3 x^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{3\epsilon^3 x^4 y^2 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{3\epsilon^3 x^2 y^4 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{\epsilon^3 y^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{\epsilon^3 x^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{3\epsilon^3 x^4 y^2 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{3\epsilon^3 x^2 y^4 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{\epsilon^3 y^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} \\
& + \frac{\epsilon^3 v x^4 k^4 \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \frac{2\epsilon^3 v x^2 y^2 k^4 \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon^2 x^4 k^2 + 2\epsilon^2 x^2 y^2 k^2 + \epsilon^2 y^4 k^2 + 2\epsilon x^2 k^2 + 2\epsilon y^2 k^2 + k^2} + \dots
\end{aligned}$$

$$\Lambda = 0$$

## Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\begin{aligned} \Psi_3 = & -\frac{i\epsilon^2 x^4 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 x^2 y^2 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 y^4 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x^4 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x^2 y^2 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 y^4 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} \\ & + \frac{i\epsilon^2 x^4 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon^2 x^2 y^2 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon^2 y^4 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x^4 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x^2 y^2 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 y^4 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{8|\epsilon x^2 k + \epsilon y^2 k + k|} \\ & + \frac{i\epsilon^2 x^2 y k^2 \frac{\partial}{\partial x} b_x(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon^2 y^3 k^2 \frac{\partial}{\partial x} b_x(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x^2 y k^2 \frac{\partial}{\partial y} b_x(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 y^3 k^2 \frac{\partial}{\partial y} b_x(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 x^3 k^2 \frac{\partial}{\partial x} b_y(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 x y^2 k^2 \frac{\partial}{\partial x} b_y(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x^3 k^2 \frac{\partial}{\partial y} b_y(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} \\ & - \frac{\epsilon^2 x y^2 k^2 \frac{\partial}{\partial y} b_y(u, x, y)}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x^2 b_x(u, x, y) k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 x y b_x(u, x, y) k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 y^2 b_x(u, x, y) k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon^2 x^2 b_y(u, x, y) k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x y b_y(u, x, y) k^2}{|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon^2 y^2 b_y(u, x, y) k^2}{2|\epsilon x^2 k + \epsilon y^2 k + k|} \\ & - \frac{i\epsilon x^2 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i\epsilon y^2 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon x^2 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon y^2 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon x^2 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i\epsilon y^2 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \dots \end{aligned}$$

[illegible]

# Newman-Penrose Equations

[illegible]

$$NP12 = \epsilon^2 x^2 \bar{k}k + \epsilon^2 y^2 \bar{k}k - \epsilon^2 x^2 k^2 - \epsilon^2 y^2 k^2$$

$$NP15 = -\frac{\epsilon^2 x^3 \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 x^2 y \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x y^2 \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^3 \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon^2 x^3 \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 x^2 y \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{\epsilon^2 x y^2 \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon^2 y^3 \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x^3 k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon^2 x^2 y k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon^2 x y^2 k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon^2 y^3 k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|} \\ - \frac{\epsilon x \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y \frac{\partial}{\partial u} k k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{\epsilon x \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} - \frac{i \epsilon y \bar{k} \frac{\partial}{\partial u} k}{4|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{\epsilon x k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|} + \frac{i \epsilon y k \frac{\partial}{\partial u} k}{2|\epsilon x^2 k + \epsilon y^2 k + k|}$$

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## Bianchi Identities

$$\begin{aligned}
BI11 = & \frac{\epsilon^3 x^4 y \bar{k} k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x^2 y^3 \bar{k} k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 y^5 \bar{k} k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^4 y k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^2 y^3 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} \\
& - \frac{\epsilon^3 y^5 k^2 \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^5 \bar{k} k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^3 y^2 \bar{k} k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x y^4 \bar{k} k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x^5 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} \\
& + \frac{\epsilon^3 x^3 y^2 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x y^4 k^2 \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^4 y \bar{k} k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^2 y^3 \bar{k} k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 y^5 \bar{k} k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} \\
& + \frac{\epsilon^3 x^4 y k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x^2 y^3 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 y^5 k^2 \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x^5 \bar{k} k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} + \frac{\epsilon^3 x^3 y^2 \bar{k} k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} \\
& + \frac{\epsilon^3 x y^4 \bar{k} k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^5 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^3 y^2 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{2(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x y^4 k^2 \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(\epsilon x^2 + \epsilon y^2 + 1)} - \frac{\epsilon^3 x^2 y^2 \bar{k} k \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{\epsilon^3 y^4 \bar{k} k \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} \\
& + \frac{\epsilon^3 x^2 y^2 k^2 \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} + \frac{\epsilon^3 y^4 k^2 \frac{\partial}{\partial x} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} + \frac{\epsilon^3 x^3 y \bar{k} k \frac{\partial}{\partial y} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} + \frac{\epsilon^3 x y^3 \bar{k} k \frac{\partial}{\partial y} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{\epsilon^3 x^3 y k^2 \frac{\partial}{\partial y} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} - \frac{\epsilon^3 x y^3 k^2 \frac{\partial}{\partial y} b_x(u, x, y)}{\epsilon x^2 + \epsilon y^2 + 1} + \dots
\end{aligned}$$

*Equations are vanished.*

## **Petrov Type**

Type "*III*"